

Trainings ▾

Remove with Mechanically Actuated Injector

With Mechanically Actuated Injectors

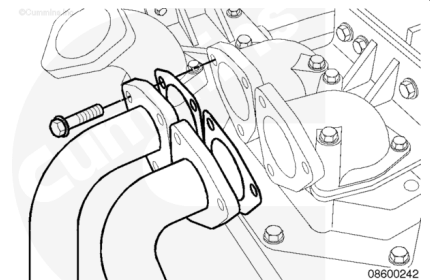
This procedure is designed to remove the two-stage assembly from a QSK60 two-stage engine with a minimal amount of disassembly and requires a minimum of 305-mm [12-in] overhead clearance.

The approximate weight, after removal of suggested components, is 885 kg [1951 lb].

It is a good practice to mark the location of all lines during removal.

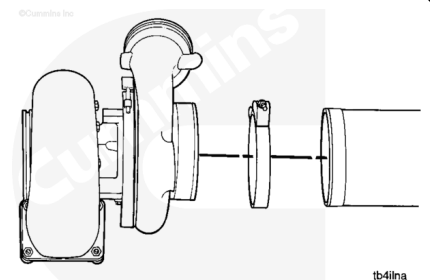
Disconnect the coolant lines from the thermostat housing.

Position the coolant lines aside.



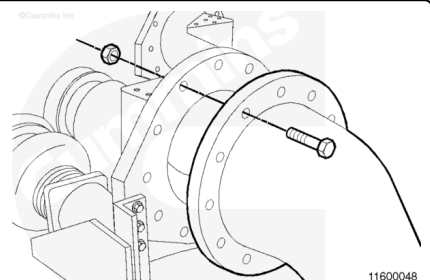
Disconnect the air intake piping between the air filter(s) and low pressure turbochargers.

Disconnect the air intake piping at the inlet side of each turbocharger.



Remove the 12 exhaust outlet flange capscrews and nuts.

Disconnect the exhaust piping from the exhaust outlet flange.

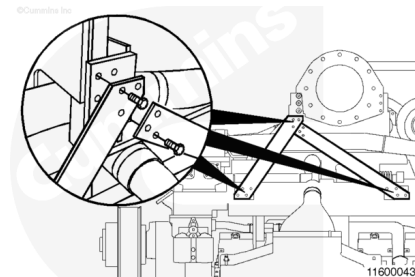


Remove the supports.

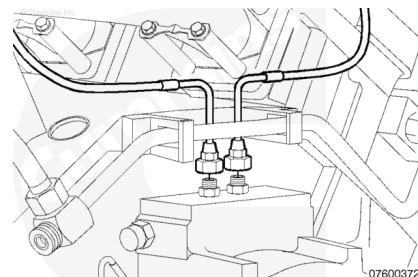
Remove the 12 capscrews connecting the air crossover to the aftercooler cover.

Remove the clamps from the hose connecting the air crossover to the high pressure turbocharger.

Remove the air crossover.

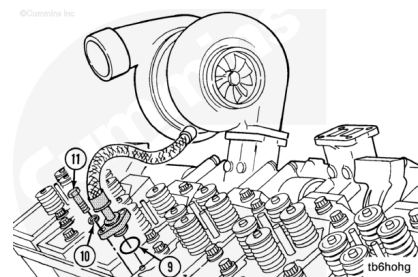


Disconnect the oil supply lines from the flywheel housing.



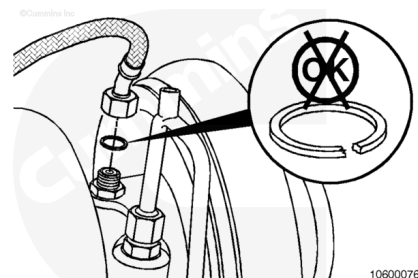
Disconnect the oil drain lines at the cylinder block for all eight turbochargers.

Remove and discard the o-ring seal (9).

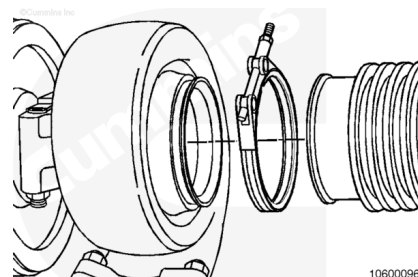


It is a good practice to mark the location of all lines during removal.

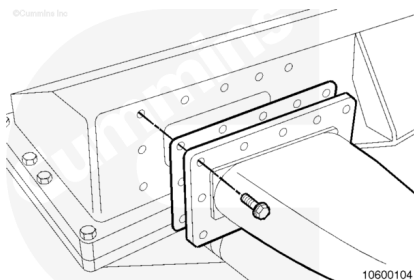
Disconnect the oil supply lines from each of the four high-pressure turbochargers.



Disconnect the V-band clamp at the bellows on the turbine side of the four high-pressure turbochargers.

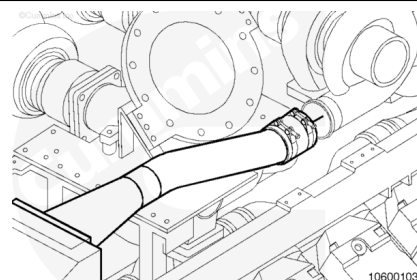


Remove the 12 capscrews from each of the four intercooler intake crossovers to the low-pressure turbochargers.



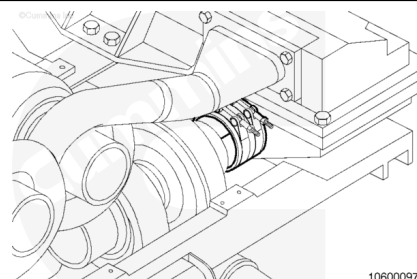
Loosen the hose clamps on each end of the four air connection hoses for the four intercooler intake crossovers and the four low pressure turbochargers.

Note : The left rear low-pressure turbocharger connection is shown. The other low-pressure turbocharger connections are similar.



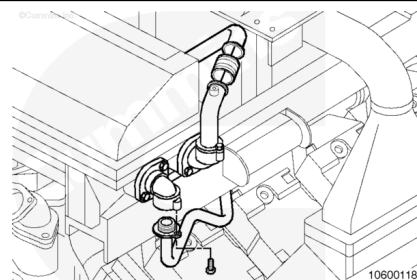
Loosen the two intake hose clamps each on the four high pressure turbochargers and the two intercoolers.

Disconnect the hose clamp at the intercooler.

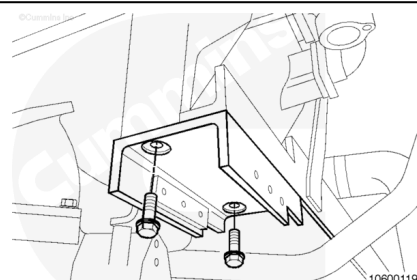


Note : The front intercooler water supply and return connections are shown. The rear intercooler connections are similar.

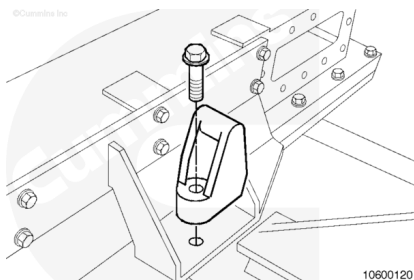
Disconnect the water supply and return piping from the intercoolers.



Remove the capscrews from each of the four feet of the intercooler.



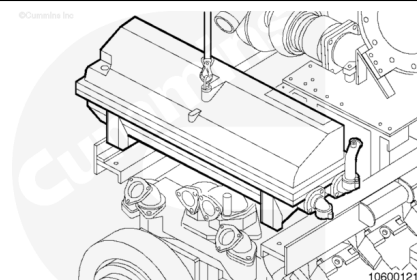
Remove the lifting eye from the bracket between the intercooler and the two-stage frame.



10600120

⚠ WARNING ⚠

This assembly weighs 23 kg [50 lb] or more. To reduce the possibility injury, use a hoist or get assistance to remove this assembly.

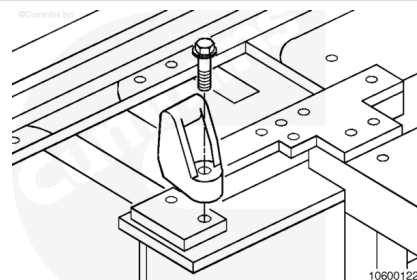


10600121

Install a 12 mm [0.47 in] lifting device in the top of the intercoolers.

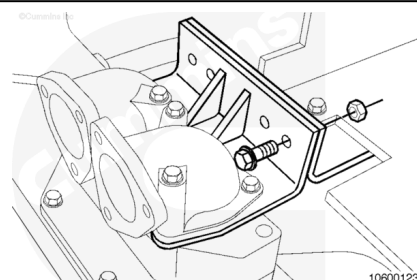
Remove the intercoolers.

Install the two lifting eyes on the two-stage frame.



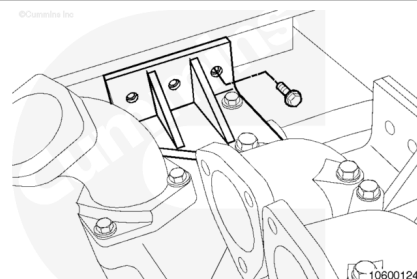
10600122

Remove the four capscrews from the bracket connecting the low temperature aftercooling housing to the two-stage frame.



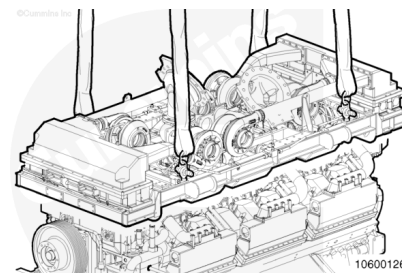
10600123

Remove the three capscrews from each of the two brackets connecting the low temperature aftercooling thermostat housing to the two-stage frame.

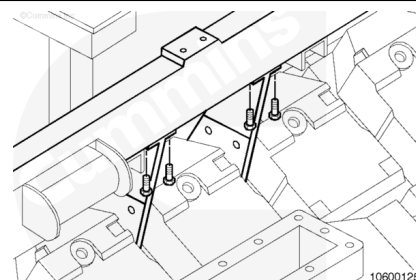


10600124

Remove the ten capscrews connecting the frame to each of the three center support brackets.

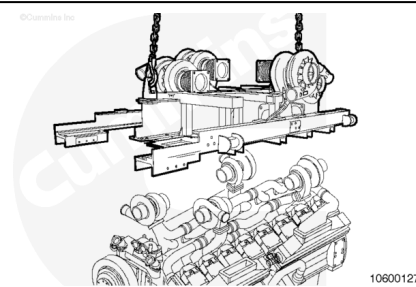


Remove the two capscrews connecting each of the outboard brackets to the two-stage frame.



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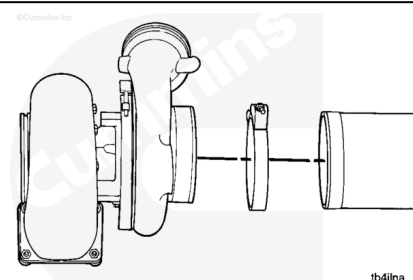
Connect the lifting straps to the two-stage frame lifting eyes.

Remove the two-stage frame from the engine.

with Electronically Actuated Injector

Disconnect the air intake piping between the air filter(s) and low pressure turbochargers.

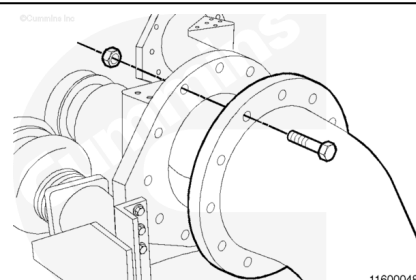
Disconnect the air intake piping at the inlet side of each turbocharger.



Remove the 12 exhaust outlet flange capscrews and nuts.

Disconnect the exhaust piping from the exhaust outlet flange.

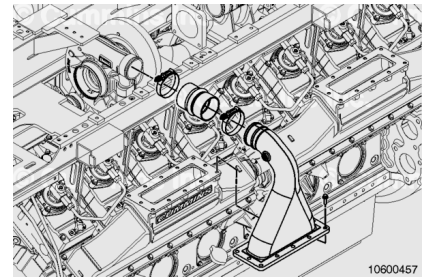
Repeat the above for the exhaust on the opposite side of the engine.



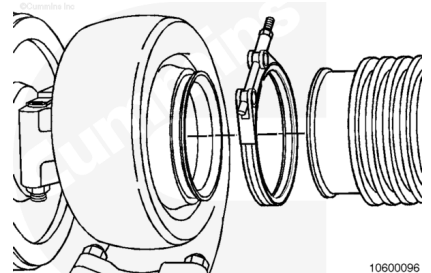
Remove the 12 capscrews connecting the air crossover to the aftercooler cover.

Remove the clamps from the hose connecting the air crossover to the high pressure turbocharger.

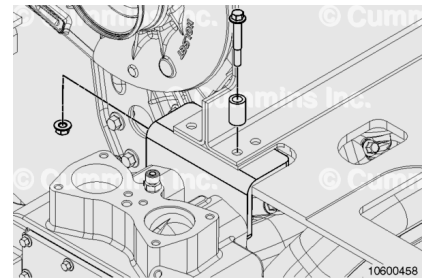
Remove the air crossover pipe.



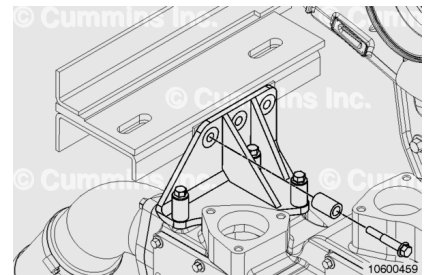
Disconnect the V-band clamp at the bellows on the turbine side of the four high-pressure turbochargers.



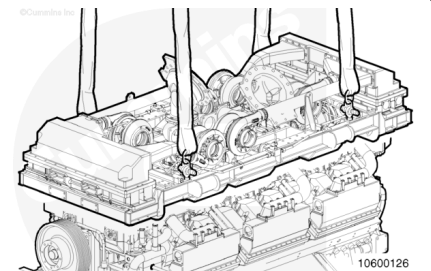
Remove the four capscrews from the bracket connecting the low temperature aftercooling housing to the two-stage frame.



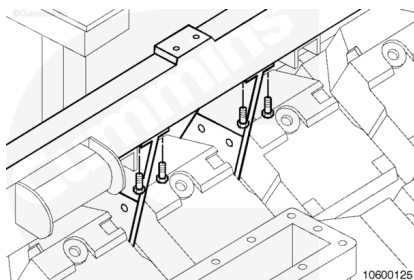
Remove the three capscrews from each of the two brackets connecting the low temperature aftercooling thermostat housing to the two-stage frame.



Remove the nine capscrews connecting the frame to each of the three center support brackets.



Remove the four capscrews connecting each of the outboard brackets to the two-stage frame.



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This assembly weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this assembly.

Connect the lifting straps to the two-stage frame lifting eyes.

Remove the two-stage frame from the engine.

